

Chapter 56 — Hypospadias — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Incidence 1 in 150–300** male births; distal 70–85%, proximal 10–25%.
- **Hypospadias + undescended testis** → DSD (disorder of sex development) work-up (karyotype, ultrasound, hormones).
- **Repair at 6–12 months** (AAP, American Academy of Pediatrics, 1996); genital awareness about 18 months.
- **Curvature <30°** → dorsal plication; **>30°** → divide plate → ventral lengthening (usually staged).
- **TIP** (tubularised incised plate): >90% of distal repairs; stop 2–3 mm short of meatus.
- **Commonest complication**: fistula (just under 10%); repair ≥6–12 months.
- **Meatal stenosis**: symptoms + <8 Fr prepubertal or <12 Fr postpubertal.
- **Buccal mucosa** = graft of choice; **lip within the glans**.
- **25–75%** need extra procedures after planned two-stage repair.
- **Follow up through puberty**: only about half of complications appear in year 1.

EMBRYOLOGY

- **Triad**: ventral meatus, ventral curvature, dorsal hood.
- **Critical window weeks 8–16**; DHT (dihydrotestosterone) key at 8–12 weeks.
- **SRY** 7–8 weeks; **5 α -reductase type 2** → DHT locally.
- **Two-zipper** (endodermal): opening zipper both sexes; closing zipper male, androgen-dependent.
- **Ventral prepuce** fuses about 20 weeks → incomplete foreskin in 95%.
- **MIP** (megameatus with intact prepuce) 5%: retract foreskin before every circumcision.

DIAGNOSIS

- **Severity**: meatus + curvature **after degloving**; distal-looking can be proximal.
- **GMS** (glans–meatus–shaft) score: each 1–4, summed; curvature <30 / 30–60 / >60°; higher = more complications.
- **Hernia or hydrocele 9–16%**; **cryptorchidism about 7%** (about 10% proximal).
- **Non-palpable testis** → DSD about 50%; proximal vs distal DSD 64% vs 7%.
- **Severe hypospadias = 46,XY DSD** (since 2006); normal karyotype can mislead.
- **Curvature without hypospadias** → defer circumcision.

AETIOLOGY

- **Risks**: prematurity (<37 weeks), SGA (small for gestational age, <10th centile), IUGR (intrauterine growth restriction), small placenta (2 $\times$), assisted reproduction.
- **Twin concordance 9–27%**; first-degree relative 9–15%; chromosomal defect up to 30%.
- **Syndromes**: WAGR, Denys–Drash, Smith–Lemli–Opitz (about 200 syndromes).
- **TDS** (testicular dysgenesis syndrome): poor semen, testis cancer, cryptorchidism, hypospadias.
- **Short AGD** (anogenital distance) $\propto$ severity; EDC (endocrine-disrupting chemical) exposure.

SURGICAL ASSESSMENT

- **No routine VCUG** (voiding cystourethrogram) or renal ultrasound (non-syndromic).
- **Stretched penile length** (micropenis) + both groins.
- **Glans <14 mm** → more complications (disputed).
- **Testosterone cypionate 2 mg/kg IM** at 5 and 2 weeks pre-op if glans <15 mm; +4 mm with two doses.
- **Androgen side effects** resolve within 6 months.
- **Caudal** commonest block; fistula link disputed; **pubdental** for older children.

- **Ketorolac 0.5 mg/kg IV** at end.
- **Cystoscopy** only for severe proximal or redo.
- **POST** (plate objective scoring tool): low AB/BC → inlay graft rather than TIP.
- **Curvature grading**: mild <30°, moderate 30–45°, severe ≥45°; goniometer.
- **Artificial erection**: tourniquet + saline (Gittes–McLaughlin); tourniquet hides basal curve.
- **Plication**: Nesbit (1965) excision; Baskin midline 12 o'clock nerve-free zone.
- **Recurrence**: plication 28% vs grafting 9%.
- **Dermal graft** 20% larger than defect; corporotomy 9 to 3 o'clock.
- **Congenital curvature** (normal urethra): recurrence 2–15%.
- **Sutures**: chromic or polyglactin > PDS (polydioxanone); 6-0 to 8-0 infants.
- **Never hair-bearing skin** for urethroplasty.
- **Belman flap** = de-epithelialised preputial dartos barrier.
- **Tourniquet** released every 30 min; dressing 72 h.

ADVANCEMENT AND TUBULARISATION

- **MAGPI** (meatal advancement and glanuloplasty incorporated; Duckett 1981): glanular meatus only.
- **MIV** (M inverted V; Decker 1991): coronal or proximal glanular; avoids retraction.
- **Urethromeatoplasty**: Heineke–Mikulicz for stenotic distal meatus.
- **GAP** (glans approximation procedure; Zaontz 1989): deep wide groove, fish-mouth meatus (MIP).
- **TD** (Thiersch–Duplay): U-incision sized to 12 Fr.
- **TIP** (Snodgrass 1994): incise plate to 12 mm total width.
- **Mathieu**: flip-flap; narrow base → fistula; SLAM (slit-like adjusted Mathieu) → slit meatus.
- **OIF** (onlay island flap): narrow plate; diverticulum if flap too wide; less stenosis.

PROXIMAL REPAIR

- **TPIF** (transverse preputial island flap, Duckett tube, 1980): oval meatus; Duckett moved to OIF.
- **OIF proximal**: plate + flap ≤12 mm.
- **Proximal TIP** abandoned (stricture, recurrent curvature).
- **Bracka / STAG** (staged tubularised autograft): free graft, 1–3 ungrafted corporotomies.
- **STAC**: three stages; avoids graft over corporotomy.
- **Byars** (1951): pedicled preputial flaps; highest staged complications.
- **Ulaanbaatar**: distal tube first; second stage at 6 months.
- **Stage interval ≥6 months** (8–12); repeat artificial erection.

- **Poiseuille:** resistance $\propto$ length, $1/r^4 \rightarrow$ long neourethra = stricture.

POSTOPERATIVE CARE AND COMPLICATIONS

- **Diversion 3–21 days;** oxybutynin for spasm; Foley must be midline.
- **Table 56.1:** curvature or torsion $>30^\circ$; glans fusion <2 mm; skin surplus ≥ 2 cm.
- **Fistula:** original meatus or corona; recurrence about 20%; coronal $\rightarrow$ revise urethroplasty.
- **Fistula work-up:** bougie à boule $\rightarrow$ urethroscopy $\rightarrow$ stress test.
- **Routine meatal dilatation** unnecessary.
- **Stricture <1 cm:** urethrotomy rarely durable.
- **Diverticulum:** preputial flap, two-stage, proximal repairs.
- **BXO** (balanitis xerotica obliterans): excise, staged buccal; recurrence up to 40%; invades buccal.
- **Utricle + recurrent UTI** (urinary tract infection) $\rightarrow$ excise or fulgurate.

OUTCOMES

- **Pfistermuller TIP meta-analysis:** complications 10.6%, reoperation 4.5%.
- **TIP: fistula 5.7% $>$ stenosis 3.6% $>$ stricture 1.3%.**
- **TIP vs Mathieu:** more stenosis with TIP.
- **Distal success 83–95%;** proximal complications **23–68%.**
- **STAG $<25\%$; Byars 30–70%.**
- **Recurrent curvature** after two-stage 0–10%.
- **Recurrent UTI after repair** about 2%.
- **HOPE** (Hypospadias Objective Penile Evaluation): photo panels; 1–10 per item.

REOPERATION

- **Devine:** no "small operations"; always artificial erection.
- **Buccal:** 10% oversize; fenestrate; wait ≥ 6 months; staged success 30–75%.
- **Bladder mucosa:** prolapse up to 38%.
- **Tissue expander:** sufficient 95%; complications 41%.
- **Cecil-Culp:** release at 6–8 weeks (Cecil) vs 9–12 months (modified).

CLASSIC TRAPS

- **Trap:** meatal position = severity — **Actually:** assess after degloving.
- **Trap:** circumcise newborn with MIP or curvature — **Actually:** retract first; save foreskin.
- **Trap:** normal 46,XY excludes DSD — **Actually:** consider 5 α -reductase and AIS (androgen insensitivity).
- **Trap:** routine VCUG — **Actually:** not indicated.
- **Trap:** plicate severe curvature — **Actually:** recurs; lengthen ventrally.
- **Trap:** tourniquet artificial erection is complete — **Actually:** hides basal curvature.
- **Trap:** MAGPI for coronal meatus — **Actually:** retraction; glanular only.
- **Trap:** excise a coronal fistula simply — **Actually:** revise the urethroplasty.
- **Trap:** meatotomy for BXO stenosis — **Actually:** excise all and stage.
- **Trap:** most complications in year 1 — **Actually:** about half; strictures late.